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1. What is Data Science? Explain its Primary Objective.

Ans Data Science is a interdisciplinary field that combines statistics, programming, machine learning and domain knowledge to extract meaningful insights from data.

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2. Why is Data Science important in today digital world?

Ans Data Science plays an important role in today's digital world. Because huge amounts of data are generated every day through social media, online transactions and smart devices.

3. List any 4 benefits of Data Science in business & Society.

Ans Data Science provides numerous benefits to business and society. It helps companies understand Customer

behaviour, improve productivity, reduce costs.

4. Mention four real-world application of Data Science.

Ans Data Science has many real-world applications across different industries. It is used in recommendation systems such as Netflix & Amazon, fraud detection in banking, disease prediction in healthcare.

5. What are different facts of data?

Ans The main facts of data are structure, source, type, time, quality & usage. Type describes the nature of data values.

6. Differentiate between Structured, Semi-structured and unstructured data.

Ans Structured data is organized in row and columns, such as database. Semi-structure data contains tags, JSON and XML. Unstructured data include image, video.

7. What is Big data? State its Key Characteristics.

Ans Big Data refers to very large and complex datasets.

Its five characteristics are volume, velocity, variety, veracity, and value.

8. Explain the Big Data ecosystem.

Ans The Big Data Ecosystem consists of tools used to store, process, and analyze large datasets. Some technologies are Hadoop, Spark.

9. What are the major phases of the Data Science Process.

Ans The major phases are data collection, data cleaning, data integration, data transformation, data analysis, and data visualization.

10. What is data retrieval? Mention two common sources.

Ans Data retrieval is the process of obtaining data from storage systems. Common sources include data bases and websites.

11. Why is data collection considered important in Data Science?

Ans: Data collection is important because accurate & relevant data leads to reliable analysis & better decision-making.

12. What is data cleaning? List any four techniques.

Ans: Data cleaning is the process of correcting error in data. Common techniques include removing duplicates, handling missing values.

13. What is data integration? Why is it necessary?

Ans: Data integration combines data from different sources, into a single dataset.

14. Explain Data Transformation.

Ans: Data transformation converts data into a suitable format for analysis.

15. What is Data Analysis?

Ans: Data analysis is the process of examine data to extract useful information.

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